

Project Report: Shrinking Core Model for Gas-Solid Reduction

Author: Prabhat Solanki

Institution: National Institute of Technology, Rourkela

1. Objective

To computationally simulate the isothermal reduction of iron ore (Fe_2O_3) pellets by carbon monoxide (CO) using the Shrinking Core Model (SCM). The analysis evaluates the influence of pellet size on the overall reduction rate and identifies the dominant resistive mechanism by calculating the mass transfer Biot number (Bi_m).

2. Methodology

The reduction process was modeled by decoupling the overall reaction time into three characteristic times representing the primary mass transfer and kinetic resistances:

- **Gas Film Diffusion:** Transport of CO gas through the stagnant boundary layer.
- **Ash Layer Diffusion:** Diffusion of CO through the porous reacted iron layer.
- **Interfacial Chemical Reaction:** The intrinsic reaction rate at the unreacted Fe_2O_3 core.

A vectorized MATLAB solver was developed to compute the conversion fraction (X) as a function of total time for standard industrial pellet radii (5 mm, 10 mm, and 20 mm). The mass transfer Biot number ($\text{Bi}_m = k_g R / D_e$) was calculated for each geometry to evaluate the ratio of internal ash diffusion resistance to external gas film resistance.

3. Results and Discussion

The computational model successfully generated conversion-vs-time profiles for the three pellet geometries. The corresponding characteristic times (τ) and Biot

numbers (Bi_m) revealed a distinct shift in the rate-controlling regime as the pellet radius (R) increased.

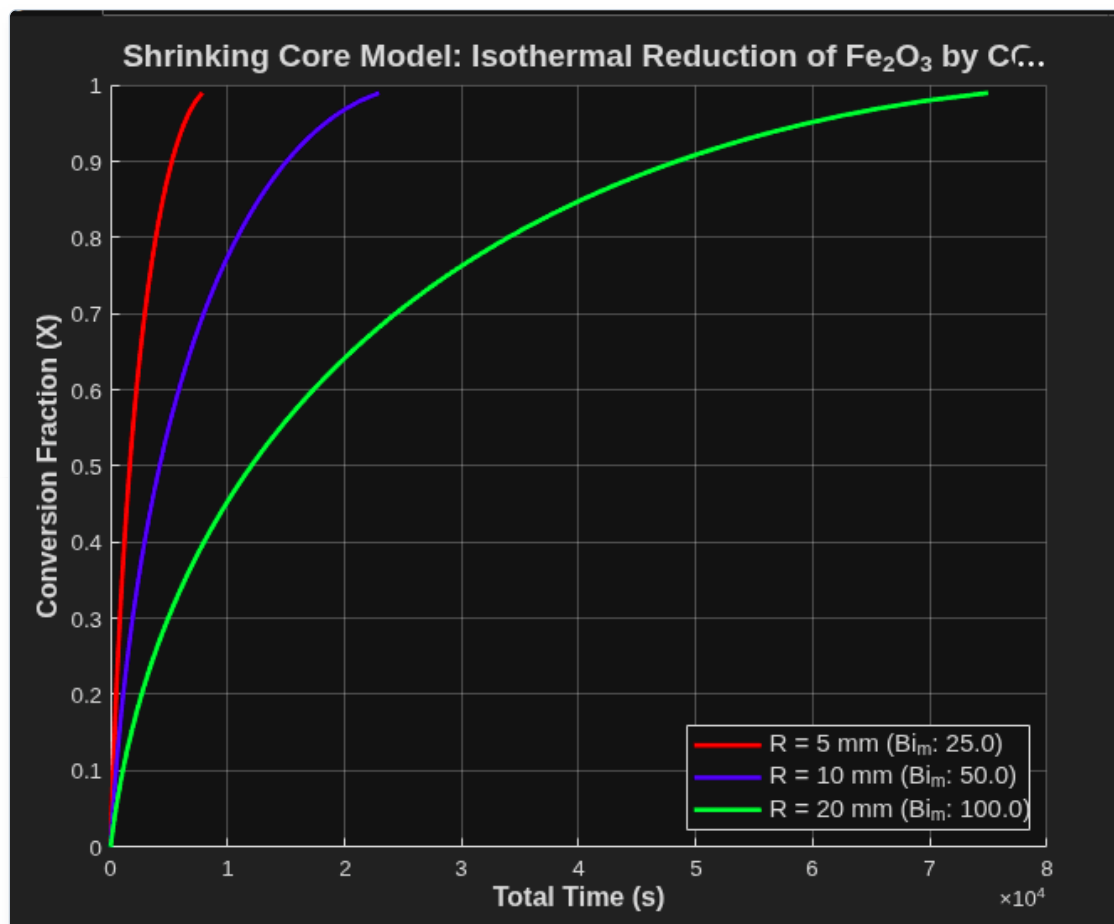


Figure 1: Conversion fraction (X) vs. Total Time for varying pellet radii showing distinct reduction profiles.

```
Command Window
>> shrinking_core_model
Radius 5 mm -> Bi_m: 25.0 | Dominant Resistance: Chemical Reaction (tau = 4950.0 s)
Radius 10 mm -> Bi_m: 50.0 | Dominant Resistance: Ash Diffusion (tau = 16500.0 s)
Radius 20 mm -> Bi_m: 100.0 | Dominant Resistance: Ash Diffusion (tau = 66000.0 s)
>>
```

Figure 2: MATLAB console output detailing dominant resistances, characteristic times, and Biot numbers for each geometry.

- **Small Pellets ($R = 5$ mm):** The mass transfer Biot number is 25.0. The dominant resistance is the interfacial chemical reaction, with a characteristic time of $\tau = 4950.0$ seconds. The conversion curve is steepest, indicating the fastest overall reduction rate.
- **Medium Pellets ($R = 10$ mm):** The mass transfer Biot number increases to 50.0. The rate-controlling mechanism shifts to ash layer diffusion, with $\tau = 16500.0$ seconds.

- **Large Pellets (R = 20 mm):** The mass transfer Biot number reaches 100.0. Ash layer diffusion strongly dominates the process, heavily throttling the reduction rate with a characteristic time of $\tau = 66000.0$ seconds. The conversion profile is significantly flattened, taking over 70,000 seconds to approach full conversion.

4. Conclusion

The model demonstrates that the controlling resistance during iron ore reduction is highly dependent on pellet geometry. At a 5 mm radius, the process is limited by the chemical reaction rate at the shrinking core. However, as the radius scales to 10 mm and 20 mm, the Biot number increases and the dominant bottleneck transitions to ash layer diffusion due to its quadratic dependence (R^2) on the pellet radius. Optimizing industrial reduction rates for larger pellets therefore requires strategies that enhance the effective diffusivity of the porous product layer rather than solely increasing gas flow or intrinsic reaction kinetics.